Preliminary Assessment of Gold, Platinum Group Elements (PGE), and Rare Earth Elements (REE) Potential

Staked Claims Area Barstow “Green Mountain” Region, San Bernardino County, California

This preliminary, high-level assessment evaluates the mineral exploration potential of the staked claims located southwest of Barstow, California, west of Hodge, and north of Helendale, in the vicinity of the Iron Mountain area, historically referred to as the Green Mountain region.

The presence of both lode and placer gold claims within and adjacent to the project area indicates a dual exploration opportunity for gold, a characteristic feature of several historic mining districts in the Mojave Desert. The regional geology is dominated by Mesozoic plutonic and metavolcanic rocks, creating a geologically complex but favorable environment for mineralization. In addition to gold, historical reports from the Green Mountain area suggest the potential occurrence of platinum group elements (PGEs) and rare earth elements (REEs) associated with mafic to ultramafic lithologies, thereby broadening the overall exploration potential of the property.

Geologic mapping of the claims area identifies three principal lithologic units relevant to mineral exploration:

- The dominant unit within the claim grid is designated “gb” and consists primarily of Mesozoic plutonic rocks of gabbroic to dark dioritic composition. Although felsic intrusions are more commonly associated with classic gold-bearing quartz vein systems, mafic intrusive rocks such as gabbro and diorite are recognized hosts for structurally controlled gold mineralization in numerous districts. Their brittle deformation response and chemical reactivity with hydrothermal fluids can facilitate vein development along faults, fractures, and shear zones, particularly near intrusive contacts.

Historical reports from the area describe mineralization within coarse-grained, ferromagnesian-rich intrusive rocks containing pyroxene, hornblende, olivine, and related minerals. These reports document occurrences of gold and silver, as well as platinum group elements—including platinum and palladium—within basic to ultrabasic rocks, likely corresponding to the gabbro unit. The PGE and REE mineralization has historically been interpreted as resulting from magmatic segregation and high-temperature processes within deep-seated intrusive bodies. Although the reported grades and extents require field verification, the association of PGEs with mafic and ultramafic lithologies is geologically plausible and consistent with global analogs.
Limited historical sampling has also reportedly identified rare earth elements, including lanthanum, cerium, and yttrium, within portions of the Green Mountain project area. These occurrences, noted in association with basic intrusive rocks and detected through spectrographic analyses, represent a potential secondary exploration target for future evaluation.

- The “GrMz” unit, comprising Mesozoic granite, quartz monzonite, and granodiorite, occurs adjacent to the gabbroic intrusions and represents a more typical host for lode gold mineralization throughout the western United States, including the Mojave Desert region. The spatial relationship between mafic (gb) and felsic (GrMz) intrusive bodies creates favorable conditions for mineralization, particularly along contacts where enhanced permeability may have facilitated hydrothermal fluid flow.

Contacts between these intrusive units, together with associated fault and fracture systems, are considered priority targets for lode-style mineralization. Any lode-hosted gold mineralization within these units could serve as a source for placer gold accumulation in surrounding alluvial systems, consistent with historical placer activity in the broader Barstow–Iron Mountain area.

- The “Mv” unit, composed of pre-Cenozoic metavolcanic rocks including latite, dacite, tuff, and greenstone, occurs more peripherally within the project area but remains a regionally prospective lithology for gold mineralization, particularly where structural preparation and alteration are evident.

Conclusions and Recommended Next Steps

The staked claims area exhibits significant mineral potential, primarily for gold, owing to its location within the Mojave Desert mineral province and the presence of Mesozoic intrusive assemblages known to host both lode and placer deposits in analogous settings. The juxtaposition of mafic and felsic intrusions supports multiple exploration models and target styles.

Historical documentation further suggests the potential presence of platinum group elements and rare earth elements within mafic to ultramafic intrusive rocks. Although these observations require modern verification, they expand the commodity scope of the project and highlight additional exploration upside beyond gold.

A phased, reconnaissance-level exploration program is recommended. Initial efforts should include detailed geological and structural mapping, systematic prospecting, and targeted rock-chip sampling within gabbro–granite contact zones, shear zones, and brittle fracture networks. Analytical suites should incorporate multi-element assays to evaluate PGE and REE potential alongside gold where geologically warranted.

For placer evaluation, preliminary stream sediment sampling or dry concentration techniques within alluvial drainages are recommended to assess downstream dispersion from potential lode sources. Positive results could justify follow-up trenching or shallow drilling.

References

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